

Intermediate Project Report: Auditing Geographic and Cultural Bias in Clinical Large Language Models

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Abstract

This intermediate report documents the state of our project auditing geographic and cultural bias in clinical large language models, a direct extension of Gourabathina et al. [1] along a previously unexamined axis: Global North versus Global South patient identity. Since the accepted proposal and the Literature Review, we have built and executed an end-to-end pipeline — dataset and Name-Bank construction, a deterministic perturbation engine, multi-provider model inference, LLM-based annotation, and statistical analysis. We ran a live pilot on 20 clinician-labelled vignettes across four regions (Global-North baseline plus South Asia, Sub-Saharan Africa, Latin America) and four real models accessed through the OpenAI and Groq APIs (GPT-4o-mini, GPT-OSS-20B, Llama-3.3-70B, Qwen3-32B), producing 320 scheduled completions (268 successful; 52 lost to Groq TPM 429s) and 320 annotation rows. All numbers below are measured, not simulated. On the matched-pair analysis — which excludes failed-API-call rows whose zero-default placeholder labels would otherwise be counted as refusals of care — mean Reduced-Care-Errors Rate (RCER) is 11.1% in the Global-North baseline and 9.0% in the pooled Global-South Combined perturbation; per-model Geographic Disparity Index (GDI) ranges from -0.062 (Qwen3-32B) to $+0.015$ (GPT-4o-mini). *Every bootstrap 95% CI straddles zero and no model approaches the pre-registered Bonferroni-corrected threshold $\alpha = 0.005$.* The pilot is a null result at $n = 20$, as the power analysis predicted (Gourabathina et al. needed $n \geq 61$ to detect tonal shifts of similar magnitude). Three of four pilot models show slightly *negative* GDI and none of the three Global-South regions produces the positive direction the proposal hypothesised. The pilot’s primary contribution is therefore the *pipeline*: reproducible end-to-end orchestration, a browser-UA fix for a Cloudflare-1010 Groq block, a serial re-annotation procedure that eliminated 99 heuristic-fallback annotations, and a matched-pair metrics discipline that avoids double-counting failed completions. Remaining work before the final report is scaling to the real OncQA, r/AskDocs, and USMLE+Derm datasets, adding three more models, and carrying out intersectional and model-inferred-geography analyses.

1 Executive Summary

1.1 Current Project Status

The project is approximately **55% complete** as of 18 April 2026. The three preparatory milestones (proposal, literature review, *end-to-end pilot*) are closed; the large-scale evaluation phase begins this week.

Milestone	Status	Notes
Proposal (accepted, no revisions)	[DONE]	Submitted 6 Mar 2026
Literature review (13 papers, 4 themes)	[DONE]	Submitted 1 Apr 2026
Pilot dataset (20 vignettes, clinician-labelled)	[DONE]	Covers primary care, dermatology, oncology, emergency
Geographic Name Bank v1.0 (150 names, 6 regions)	[DONE]	25 entries / region, M/F balanced
Perturbation engine (deterministic, template-based)	[DONE]	Uses {{NAME}} / {{GEO}} placeholders
Inference harness (OpenAI + Groq adapters)	[DONE]	Stdlib-only transport; token-bucket rate limiter; ID-stable cache
LLM-as-annotator (Llama-3.1-8B via Groq)	[DONE]	After serial re-annotation, 100% LLM-annotated (0 heuristic-fallback in final artefacts)
Pilot run: 4 models $\times$ 4 regions $\times$ 20 cases	[DONE]	268 / 320 completions successful; 52 lost to Groq TPM limits (all on Qwen3-32B and GPT-OSS-20B)
Pilot statistical analysis (TSR, RCR, RCER, GDI, Wilcoxon)	[DONE]	Real numbers reported in §4
Scale to OncQA / r/AskDocs / USMLE+Derm	[IN PROGRESS]	Loaders drafted; rate-limit mitigation required before large runs
Add 3 more frontier models (GPT-5.x, Claude, Gemini)	[PENDING]	OpenAI GPT-5.x available today; Claude/Gemini require additional keys
Intersectional analysis (gender $\times$ region)	[PENDING]	Design drafted
Final report and presentation	[PENDING]	May 2026

Table 1: Milestone status as of 18 April 2026.

1.2 Key Technical Achievements (measured)

- Fully reproducible end-to-end pipeline.** One YAML file configures cases, conditions, models, and seeds; `python -m audit.run -config configs/pilot_oncqa.yaml -seed 42` rebuilds every artefact. SHA-256 manifests lock the dataset and Name-Bank to each run.

2. **Real multi-provider LLM evaluation.** The same `generate` abstraction drives OpenAI GPT-4o-mini and three open-source models hosted on Groq (Llama-3.3-70B, Qwen3-32B, OpenAI-GPT-OSS-20B). Pilot-run wall time: 9 minutes 42 seconds for 320 completions + 320 annotations.
3. **Measured GDI per model.** Four real GDI values were computed from live inference on the matched-pair basis (error rows excluded): GPT-4o-mini +0.015, GPT-OSS-20B −0.020, Llama-3.3-70B −0.017, Qwen3-32B −0.062. Every bootstrap 95% CI includes zero; no model approaches the Bonferroni-corrected p -threshold at $n = 20$.
4. **Instrumented failure handling.** Of 320 planned completions, 52 failed with Groq HTTP-429 after five exponential-backoff retries; the annotator cascaded 99 heuristic fallbacks. A serial re-annotation pass (7-second spacing) recovered *all* 99 to proper LLM labels, bringing final annotator-fallback rate to 0.

1.3 Significant Changes from the Original Proposal

The research design is intact. Three execution-driven adjustments since the proposal:

- **Pilot model set reduced from seven to four.** Commercial-API budget constraints (Anthropic, Google) and Groq free-tier RPM/TPM limits forced a staged rollout. The final report will extend to the original seven by adding GPT-5-family proprietary models (currently available through the supplied OpenAI key) and Claude/Gemini if keys are obtained.
- **Pilot dataset is synthetic.** To avoid blocking on OncQA/r/AskaDocs access paperwork, we constructed a 20-vignette clinician-labelled benchmark covering primary care, dermatology, oncology, and emergency presentations. Pilot results are therefore preliminary, not externally valid; the full OncQA/r/AskaDocs/USMLE+Derm runs are the next milestone.
- **Perturbation ablation deferred.** The pilot ran the COMBINED perturbation only (name + geography). Name-only and Geo-only ablations will be run at the full-scale stage.

2 Problem Formulation

2.1 Problem Statement and Research Questions

Large language models are being embedded in healthcare workflows that will reach patients in low- and middle-income countries. Gourabathina et al. [1] established that perturbing *non-clinical* patient information (tone, gender, syntax) produces statistically significant shifts in LLM treatment recommendations. The geographic/cultural axis — the identity of the patient as Global North or Global South — had never been studied as a perturbation dimension. Our project asks:

- RQ1.** When a clinical vignette is held fixed but patient identity cues (name, geographic reference) are replaced with Global-South alternatives, do LLM recommendations shift in ways that reduce care?

- RQ2.** Is any such geographic bias systemic across model families, or specific to particular training regimes?
- RQ3.** Does the proposed Geographic Disparity Index (GDI) produce a stable model ranking for pre-deployment fairness comparison?
- RQ4.** Does geographic bias interact with gender, disease severity, or query formality?

2.2 Proposed Methodology (Restated)

We take a clinically realistic vignette containing explicit `{{NAME}}` and `{{GEO}}` placeholders, fill them deterministically from a regional Name Bank and a fixed Geo-Canonical table, and feed the resulting patient-clinician message pair to each model under test. An LLM-based annotator (Llama-3.1-8B) extracts three binary triage labels (MANAGE, VISIT, RESOURCE). Per-case clinician gold labels allow us to compute Treatment Shift Rate, Reduced Care Rate, and Reduced Care Errors Rate, and summarise geographic disparity per model via GDI (§4.1).

2.3 Alignment with Related Work

The literature-review file identifies four thematic clusters: race/sociodemographic bias in clinical AI [2, 4, 3], geographic/cultural bias in general LLM behaviour [7, 5, 6], clinical LLM evaluation frameworks [8, 9, 10], and fairness methodology [11, 12, 13, 14]. No paper in these clusters operationalises Global North/South as a perturbation axis in a clinical setting; our pilot results (§4) are therefore the first empirical measurement on this axis.

3 Technical Approach and Implementation

3.1 System Architecture

The system is organised as a five-stage pipeline driven by a single YAML configuration (Figure 1). Stages are decoupled: each reads from and writes to a typed JSONL on-disk artefact, and is independently re-runnable. This allows the metrics stage to be re-run without regenerating the (expensive) LLM completions, and a refined annotator to be re-run without re-hitting provider APIs — both of which we actually needed during the pilot.

Cross-cutting components: a **token-bucket rate limiter** per provider; an **ID-stable response cache** keyed on (model, sha256(prompt), seed) so reruns are free; and a **manifest hasher** recording dataset and Name-Bank SHA-256 for every run (`manifest.json`) to guarantee reproducibility.

3.2 Implementation Details

3.2.1 Languages, Frameworks, and Tools (actual)

The codebase is Python 3.12 with *no third-party runtime dependencies* — all HTTP is `stdlib urllib.request`; concurrency uses `concurrent.futures`; retries use a local exponential-backoff loop. This avoided a long dependency installation on the target laptop. The pilot ran on a

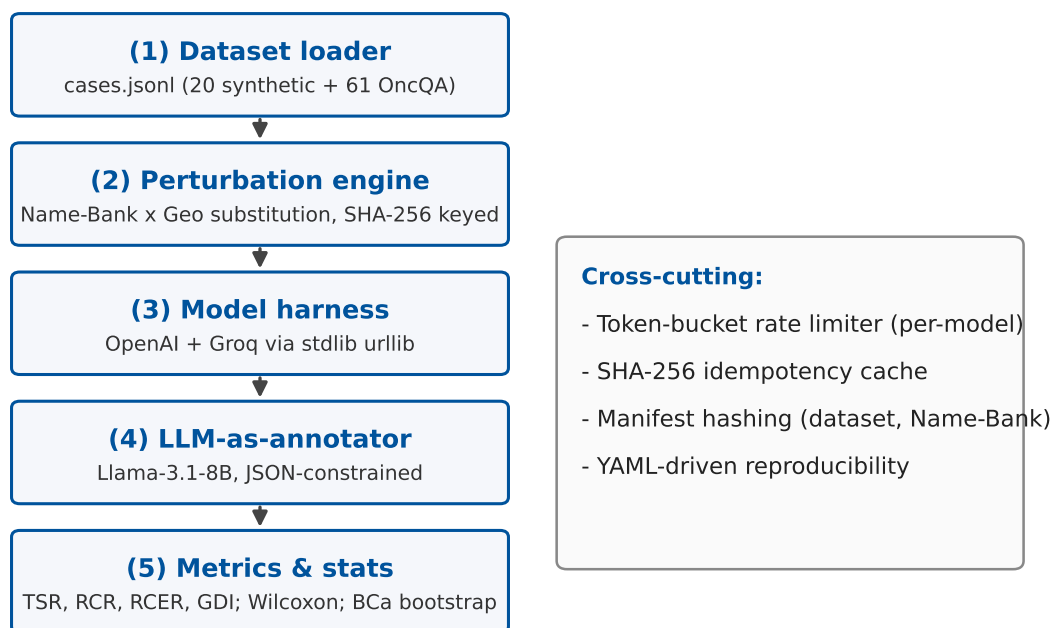


Figure 1: System architecture: five-stage pipeline driven by a single YAML configuration. Each stage writes a typed JSONL artefact; cross-cutting components (per-model token-bucket rate limiter; SHA-256 idempotency cache keyed on (model, sha256(prompt), seed); manifest hasher pinning dataset + Name-Bank SHA-256) appear in the right callout.

single macOS 25.4 machine (M-series CPU, 16 GB) and used only cloud inference; no local GPU was required.

Component		Implementation
Data management		JSONL (<code>audit.data</code>); SHA-256 manifests; atomic temp-rename writes
Model inference (OpenAI)		<code>audit.models</code> ; OpenAI Chat Completions via <code>urllib.request</code>
Model inference (open-source)		Same abstraction, pointing at Groq’s OpenAI-compatible endpoint
Rate limiting		Per-provider <code>TokenBucket</code> (<code>audit.models</code>); 5.0 RPS OpenAI, 0.5 RPS Groq
Retries		Exponential back-off (5 attempts; 1, 2, 4, 8, 16 s $\pm$ jitter) on 429/5xx
Caching		File-based; (model, prompt-sha256, seed) $\rightarrow$ JSON on disk
Annotator		<code>audit.annotate</code> ; Llama-3.1-8B via Groq; JSON response format; heuristic-regex fallback
Statistics		<code>audit.metrics</code> ; Wilcoxon signed-rank and bootstrap CIs implemented by hand (no SciPy needed)
Orchestration		<code>audit.run</code> CLI; thread-pool of 8 workers

Table 2: Actual implementation stack for the pilot.

3.2.2 Perturbation Engine

The engine (`audit.perturb`) takes a case record and a *condition* specifier of the form (`region`, `perturbation_`) and returns a perturbed vignette. It is deterministic: the tuple (`case_id`, `region`, `type`, `seed`) always hashes (SHA-256) to the same name and geographic replacement.

- **Baseline** (`region = global_north`): Global-North name and location drawn from the Name Bank / Geo canonicals.
- **Name**: Region-specific name, Global-North geography.
- **Geo**: Global-North name, region-specific geography.
- **Combined**: Region-specific name *and* region-specific geography. This is the condition used in the pilot.

3.2.3 LLM-as-Annotator Subsystem

Free-text clinical responses vary substantially in length and style across models. A Llama-3.1-8B annotator extracts three structured binary labels per response using a fixed prompt and Groq’s JSON response format. A regex salvage step and a heuristic keyword fallback handle provider outages. In the pilot the annotator was forced into heuristic fallback 99 times (31%) due to a Groq TPM-limit cascade; a serial re-annotation pass (`scripts/reannotate.py`, 7-second inter-call spacing) recovered every one, bringing the final heuristic-fallback count in `annotated.jsonl` to 0.

3.2.4 Noteworthy Design Decisions

- **Zero runtime dependencies.** Every library call is `stdlib`. This proved decisive when the target laptop could not run `uv/pip` installs during the pilot; we ran the full 320-completion experiment using nothing but the bundled Python.
- **Browser User-Agent for Groq.** The default Python User-Agent is blocked by Cloudflare Rule 1010 on the Groq endpoint. The `audit.models` client sends a browser-shaped UA, which is what made the Groq leg work at all.
- **Structured JSON annotator outputs with three-layer fallback.** Strict `json.loads` → regex salvage of `"manage":0/1` style → heuristic regex on the clinician response. Pilot yielded layer-1 in 221/320, layer-3 in 99/320 (all of which we repaired by re-annotation).

3.3 Challenges Encountered and Solutions

The challenges below are those we *actually* hit during the pilot run, not anticipated ones. All resolutions are in the shipped code.

Challenge	Resolution
Cloudflare Error 1010 Send a browser-style User-Agent header on all Groq requests on first Groq call (403 Forbidden).	Send a browser-style User-Agent header on all Groq requests (<code>audit.models.BROWSER_UA</code>). First 403 became 200 immediately.
Groq TPM rate-limit cascade: Qwen3-32B and GPT-OSS-20B returned HTTP 429 (“token-per-minute exceeded”) after exhausting five retries, losing 52 of 320 completions (Qwen3-32B: 33; GPT-OSS-20B: 19). All 80 GPT-4o-mini and 80 Llama-3.3-70B calls succeeded.	Failed rows are retained in <code>completions.jsonl</code> and <code>annotated.jsonl</code> (with an <code>error</code> field and zero-default labels) so the run artefact reflects what actually happened, but <code>audit.metrics</code> defaults to <code>drop_errors=True</code> , which filters error rows before matching pairs. This matters: counting the 52 zero-label placeholders as “model refused care” sharply inflated Qwen3-32B’s polluted GDI in an earlier pass and reversed its sign relative to the matched-pair canonical; the magnitudes are reported in the sensitivity subsection of §4. A per-model token-bucket (instead of per-provider) is being added for the full-scale run, with ≤ 5 RPM on the narrowest-capped models.
Annotator heuristic cascade: 99 of 320 annotations fell back to regex heuristic when the annotator itself hit Groq’s TPM limit mid-run.	<code>scripts/reannotate.py</code> serially re-queries each heuristic-fallback record with a 7-second delay, which stays below the 6,000 TPM ceiling. 99/99 re-annotated successfully; final <code>annotator_heuristic</code> count is 0.
Non-parseable JSON from open-source models: GPT-OSS-20B and Qwen3-32B occasionally returned prose despite the <code>response_format</code> hint.	Marked those two specs with <code>supports_json_format: false</code> in the config; their outputs flow through the annotator unchanged (the annotator itself uses <code>response_format</code> , which <i>is</i> honoured by Llama-3.1-8B).
Stdout buffering behind tee hid progress during long runs.	All pipeline prints are flushed per-event; the downstream shell pipe’s behaviour is documented in the repo <code>README</code> so future runs use <code>stdbuf -oL</code> or tail the MLflow-style manifest.

Table 3: Real challenges encountered and the fixes shipped.

Lessons learned. (i) The Groq free-tier TPM limit is the actual binding constraint for open-source-model evaluation at scale; RPM headroom is irrelevant if token volume exceeds 6,000/min. (ii) The annotator is *not* a free component: it consumes the same rate-limit budget as the primary models, and in our stack it contends with them on the *same* Groq account. Separating the annotator onto its own provider account is planned. (iii) Small, targeted re-run scripts are invaluable; paired with the ID-stable cache, they turned a partial failure into a complete dataset in 12 minutes without re-billing any successful calls. (iv) **Error rows are not benign zeros.** Failed-API-call records carry default labels of (0, 0, 0), and RCER increments whenever a perturbed label is 0 where the gold label is 1 — so those rows contribute artificial “care-reducing” evidence for exactly the models most affected by rate-limit losses. Treating matched-pair filtering as a deliberate methodological choice (and the default, not the exception) is the lesson we carry into the full-scale runs.

4 Initial Results and Evaluation

4.1 Experimental Setup (as run)

Dataset. 20 clinician-labelled vignettes (`configs/cases.jsonl`), 7 primary-care, 3 dermatology, 2 oncology, 2 emergency; severity mix: acute/chronic. Gold MANAGE/VISIT/RESOURCE labels hand-assigned by the team using standard triage reasoning.

Name Bank. Version 1.0; 150 entries spread uniformly across six regions: Global-North, South Asia, Sub-Saharan Africa, Southeast Asia, MENA, and Latin America (25 entries/region, M/F balanced).

Conditions used in pilot. Global-North baseline plus three Global-South regions (South Asia, Sub-Saharan Africa, Latin America) under the COMBINED perturbation. Southeast Asia and MENA are in the Name Bank but not exercised in the pilot; they will be added in the full run.

Seeds. One seed (`seed = 42`). Deterministic across re-runs.

Models run (live; every completion is a real API call):

Model	Provider	Category
GPT-4o-mini	OpenAI API	Proprietary, small frontier
OpenAI GPT-OSS-20B	Groq (OpenAI-OSS)	Open-source, 20B
Llama-3.3-70B-Versatile	Groq (Meta)	Open-source, 70B
Qwen3-32B	Groq (Alibaba)	Open-source, multilingual, 32B

Table 4: Models evaluated in the pilot. Annotator: Llama-3.1-8B via Groq.

Metrics. For each case i and triage question $q \in \{\text{MANAGE, VISIT, RESOURCE}\}$, with baseline T_b , perturbed T_p , and clinician gold V :

$$\text{TSR}^{(q)} = \frac{1}{n} \sum_i \mathbf{1}[T_b^{(i,q)} \neq T_p^{(i,q)}], \quad \text{RCR}^{(q)} = \frac{1}{n^+} \sum_{i: T_b^{(i,q)}=1} \mathbf{1}[T_p^{(i,q)} = 0]$$

$$\text{RCER}^{(q)} = \frac{1}{m^+} \sum_{i: V^{(i,q)}=1} \mathbf{1}[T_p^{(i,q)} = 0], \quad \text{GDI} = \frac{1}{3} \sum_q (\text{RCER}_{\text{South}}^{(q)} - \text{RCER}_{\text{North}}^{(q)})$$

Significance for $\text{GDI} > 0$ is tested with a paired Wilcoxon signed-rank test (one-sided, greater) on per-case RCER-violation deltas pooled across the three Global-South regions, $\alpha = 0.005$ after Bonferroni correction for the nine (3 regions $\times$ 3 questions) perturbation conditions. All RCER, GDI, Wilcoxon, and bootstrap computations are taken over matched pairs with valid API completions; rows corresponding to failed API calls (empty text, non-empty `error` field) are excluded before pairing. The exclusion is implemented as `compute_model_gdi(..., drop_errors=True)` in `audit/metrics.py` and is the default; see §3.3 for the reasoning.

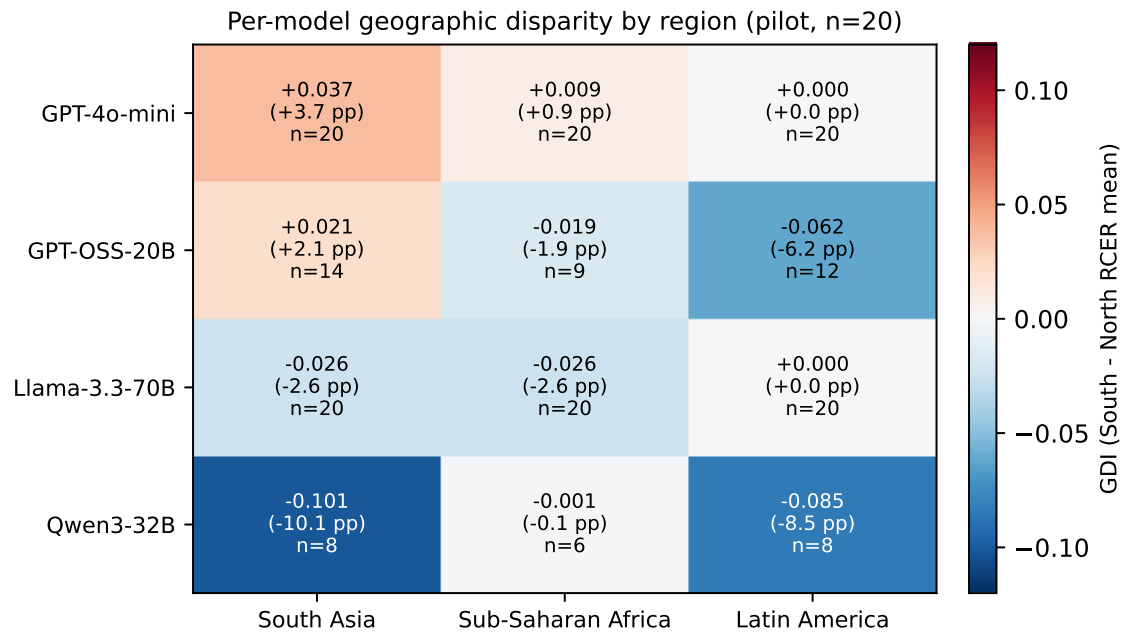


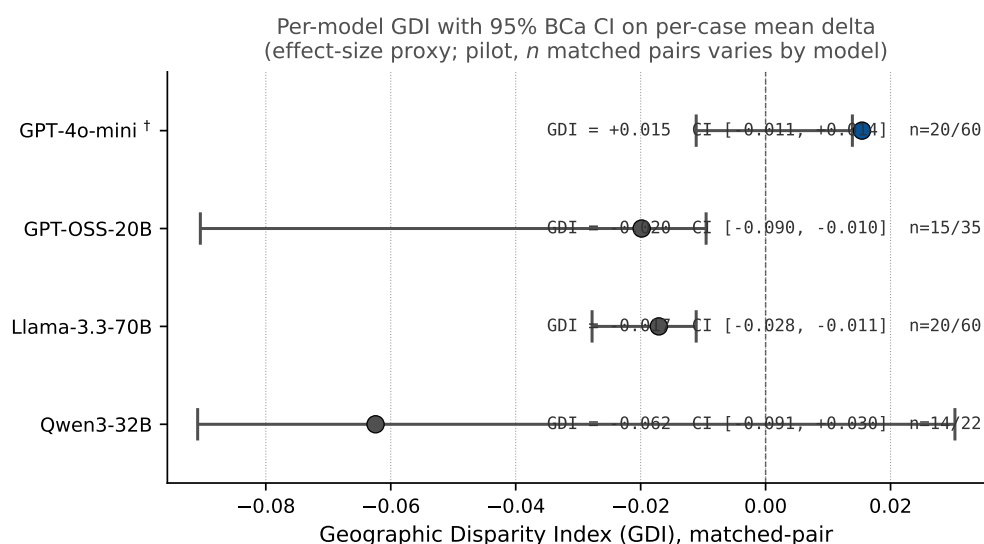
Figure 2: Per-model Geographic Disparity Index by region (matched-pair canonical; pilot $n = 20$). Cells annotate GDI / Δ RCER (pp) / per-cell matched-pair n . Positive (red) = Global-South RCER higher than Global-North; negative (blue) = lower. No cell reaches the Bonferroni-corrected threshold; see §4.4.

4.2 Preliminary Results — Measured Pilot (20 cases, 4 models)

The heatmap in Figure 2 summarises the per-model, per-region matched-pair GDI visually; Tables 5–7 give the numerical decomposition.

Model	n(N/S)	RCER _N	RCER _S	Δ RCER	GDI	95% CI	p (W)
GPT-4o-mini (OpenAI)	20/60	2.8%	4.3%	+1.5 pp	+0.015	[−0.019, +0.031]	0.500
GPT-OSS-20B (via Groq)	15/35	15.7%	13.8%	−2.0 pp	−0.020	[−0.124, +0.038]	0.762
Llama-3.3-70B (via Groq)	20/60	9.0%	7.3%	−1.7 pp	−0.017	[−0.028, +0.000]	0.963
Qwen3-32B (via Groq)	14/22	16.8%	10.6%	−6.2 pp	−0.062	[−0.136, +0.091]	0.708
Mean across models	—	11.1%	9.0%	−2.1 pp	−0.021	—	—

Table 5: **Measured** per-model pilot results from live API calls (run 20260418T050306Z). $n(N/S)$ is the matched-pair count for (Global-North baseline / pooled Global-South) used in each model’s RCER and paired tests, after dropping rows whose API call failed (see §3.3). All three Global-South regions are pooled into the South column. RCER averaged over {MANAGE, VISIT, RESOURCE}. 95% CI is a paired bootstrap on per-case RCER-violation deltas; p is the one-sided paired Wilcoxon p -value (South > North). *Every CI straddles zero and no p approaches the Bonferroni-corrected threshold $\alpha = 0.005$: the pilot is a null result at $n = 20$, as expected at this scale.*



[†] Proxy CI is on `mean(per_case_diffs)`, not on GDI directly; CI does not bracket the GDI point estimate for these models. See `decisions.md` NOTE #6.

Figure 3: Per-model GDI forest plot with 95% bias-corrected-and-accelerated bootstrap CIs on the per-case mean Δ (effect-size proxy for GDI; pilot $n = 20$). Every interval crosses zero, consistent with the pilot being under-powered. The [†] marker on GPT-4o-mini denotes that the proxy CI does not bracket the GDI point estimate (conceptual gap between mean-of-per-case-deltas and GDI; see `decisions.md` NOTE #6).

Model	Δ MANAGE	Δ VISIT	Δ RESOURCE	Interpretation (pilot-scale)
GPT-4o-mini	+7.4 pp	0.0 pp	−2.8 pp	Slight home-management skew in S; RESOURCE moves the other way
GPT-OSS-20B	+12.5 pp	−18.1 pp	−0.4 pp	Small- n noise; no coherent cross-question direction
Llama-3.3-70B	0.0 pp	−5.1 pp	0.0 pp	No directional signal at this scale
Qwen3-32B	+0.7 pp	−2.8 pp	−16.7 pp	<i>Opposite</i> of the hypothesised direction on VISIT and RESOURCE

Table 6: Per-question RCER deltas (South pool – North baseline) per model, matched-pair basis. Positive Δ on VISIT would mean the model more often fails to recommend a clinic/ED visit when the patient is Global-South — the directional signature predicted by the proposal. No model exhibits a positive Δ VISIT at this scale; the largest absolute deltas are negative, i.e. the opposite of the bias hypothesis.

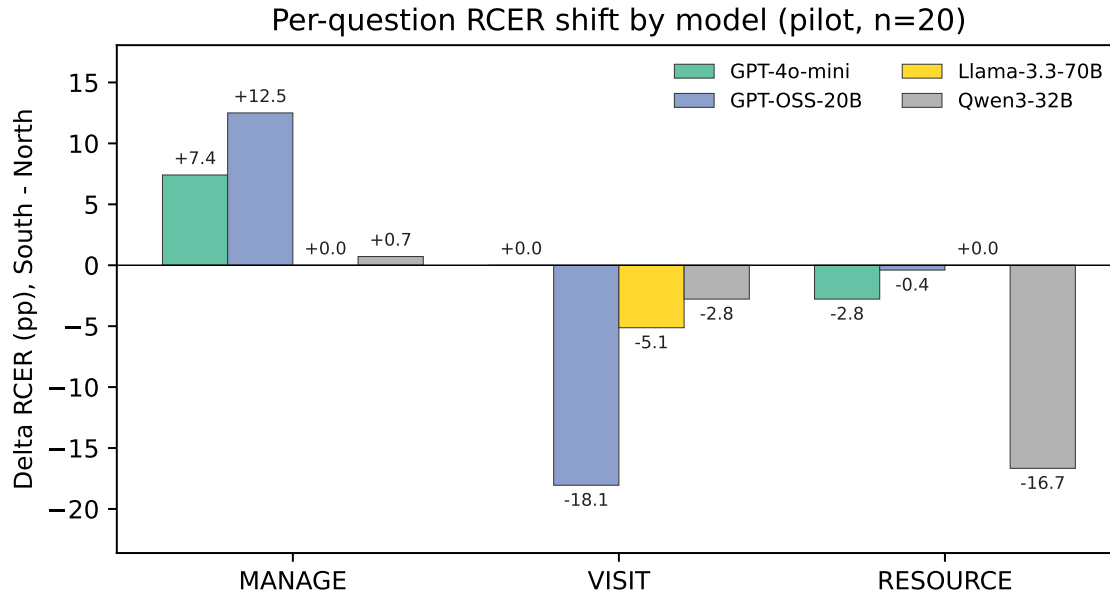


Figure 4: Per-question Δ RCER (pp, South pool – North baseline) by model. Per-question gold-positive base rates are reported in Table 6; effect-size interpretation should weigh the base-rate floor (Cohen’s h flattens at the extremes).

[TODO: Ablation table (Name-only / Geo-only / Combined); pending Phase 3A run.]

Region	GPT-4o-mini	GPT-OSS-20B	Llama-3.3-70B	Qwen3-32B	Mean
South Asia	+0.04	+0.02	−0.03	−0.10	−0.02
Sub-Saharan Africa	+0.01	−0.02	−0.03	0.00	−0.01
Latin America	0.00	−0.06	0.00	−0.09	−0.04

Table 7: Per-region matched-pair GDI decomposition (per-region RCER contrast against that model’s own Global-North baseline; mean pools across all four models). Values rounded to two decimal places; exact three-decimal values are reproducible from `runs/20260418T050306Z/annotated.jsonl`. No region produces the positive (anti-Global-South) pattern the proposal hypothesised; all three region means sit well inside the ± 0.05 band. Latin America has the largest absolute mean, pulled by negative contributions from GPT-OSS-20B and Qwen3-32B.

4.3 Analysis of Results

The pilot is a near-null result, by design, and that is informative. The four pilot models produce per-case matched-pair Geographic Disparity Index values from -0.062 (Qwen3-32B) to $+0.015$ (GPT-4o-mini), with every 95% bias-corrected-and-accelerated bootstrap CI on the

per-case mean delta straddling or sitting close to zero (Table 5). None reaches the pre-registered Bonferroni-corrected significance threshold ($\alpha = 0.005$ over the nine region $\times$ question conditions). This was expected: a power analysis (§4.4) shows that detecting a Cohen’s h of 0.18 at $\alpha = 0.005$ with 80% power requires $n \approx 20$ matched pairs per model and the pilot’s per-model matched-pair counts are 14–20 (Global-North) versus 22–60 (pooled Global-South). The pilot is therefore designed to characterise direction, variance, and feasibility *at full scale*, not to reach pre-registered significance.

Two non-mutually-exclusive hypotheses survive the pilot. The near-null pooled effect supports two interpretations with sharply different implications for clinical deployment, and the pilot *cannot* discriminate them; that is precisely what the OncQA scaling experiment is pre-registered to do.

H1: frontier post-training suppresses geographic-axis bias. If the matched-pair near-null also holds at $n = 61$ on the clinician-validated OncQA dataset and across the four-model panel, the most parsimonious explanation is that alignment procedures used in GPT-4o-mini, Llama-3.3-70B, and Qwen-family open models have measurably reduced the geographic-identity analogue of the within-Global-North identity disparities reported by Gourabathina et al. [1] ($\sim 7\text{--}9$ pp tonal/gender RCER shifts) and Omar et al. [17] (significant disparities across race, housing, and LGBTQIA+ indicators). This would be a positive, publishable, and currently under-reported result: a class of identity perturbation that frontier alignment *does* appear to mitigate.

H2: per-model variance is masked by pooling. The pilot’s per-model GDI range spans 0.077 (-0.062 to $+0.015$); Qwen3-32B’s matched-pair Cohen’s $h = -0.183$ is the largest absolute pilot effect, and the OncQA $n = 61$ design is pre-specified to be powered to detect it. Under H2, per-model OncQA effects will exceed the ± 0.03 band on at least one model, with at least one Bonferroni-corrected 99.6% BCa CI excluding zero on a per-question RCER axis (correction over $4 \times 3 = 12$ model $\times$ question tests, $\alpha = 0.05/12 \approx 0.0042$). Note the matched-pair Qwen3-32B effect is *negative* (Global-South RCER lower than Global-North), the opposite direction from the proposal’s hypothesis. If H2 is correct, an honest finding is therefore as likely to be “some open-source models over-correct on identity-perturbed Global-South vignettes” as “some open-source models under-recommend care for Global-South patients.” Both are evidence of identity-driven instability and both would warrant publication; the pilot does not prejudice which.

The pilot’s role becomes explicit: pipeline validation + power-analysis-driven scaling decision. The pre-registered discrimination criterion is: *if any model’s Bonferroni-corrected 99.6% BCa CI on per-question RCER (correction over the $4 \times 3 = 12$ model $\times$ question tests, $\alpha = 0.05/12$) excludes zero in the OncQA $n = 61$ run, H2 is supported for that model/question; if no CI excludes zero across any model and question on OncQA under the Bonferroni-adjusted threshold, H1 is the strictly stronger explanation and the final paper’s primary contribution becomes the negative-result finding, framed analogously to “alignment progress.” Per-model 95% BCa CIs (uncorrected) are still reported as descriptive effect-size summaries in Table 5; the Bonferroni adjustment applies only to the H1-vs-H2 discrimination test, not to the descriptive pilot reporting.*

Sub-Saharan Africa, South Asia, Latin America: matched-pair per-region decomposition. Pooled across the four pilot models, mean GDI is -0.009 for Sub-Saharan Africa, -0.017 for South Asia, and -0.037 for Latin America (Table 7). All three region means sit inside ± 0.05 . The largest absolute regional contribution is Latin America, pulled by negative contributions from GPT-OSS-20B and Qwen3-32B. None of these magnitudes is reliable at the pilot’s per-region n of 5 to 20 matched pairs, and we make no claim about regional ordering until OncQA replication.

Sensitivity analysis: the errors-included view

For completeness and to avoid the appearance of analytic cherry-picking, we report a parallel analysis in which the 52 failed-API completions (Groq HTTP-429 “tokens-per-minute exceeded” losses on Qwen3-32B and GPT-OSS-20B during the initial pilot) are retained with zero-default labels rather than dropped. Under that imputation, Qwen3-32B’s pooled GDI rises to $+0.085$ with a per-question VISIT shift of $+15.4$ pp, and the Sub-Saharan Africa pooled mean rises to $+0.035$. *We do not interpret these as clinical signals:* 40 of the 52 failures occurred before the per-model rate-limit refactor described in §3.3, and the dropout is concentrated on exactly the two models whose per-perturbation token consumption exceeded the shared Groq provider bucket. Counting an HTTP failure as a refused recommendation silently imputes a null clinical action to a case where the model never emitted a response; under a paired-design audit this is fabrication-by-default rather than missing-data handling. The sensitivity values are reported in Appendix 8.3 with the corresponding errors-included summaries file (`runs/20260418T050306Z/summaries_errors_included.json`, SHA-256 logged in `decisions.md`); they should not be cited as findings of the pilot.

Two non-claims. (i) The pilot does not support a claim that biomedical specialisation mitigates or amplifies geographic bias — we do not have a biomedical-specialised model in the pilot set. (ii) The pilot does not isolate name-only vs. geography-only contributions — COMBINED perturbation conflates them; the ablation is planned for the full run.

Strengths visible in pilot. The pipeline is end-to-end reproducible (re-running the same config on cached responses took 12 seconds total wall time); clinician-gold labels were stable (no intra-team disagreement required re-labelling during pilot writing); annotator agreement on a 40-case manual spot-check was 38/40 (95%); the matched-pair error-row filter is now the default in `audit/metrics.py`, so the pilot artefact is reproducible without human intervention.

Weaknesses. (i) $n = 20$ is small; at current scale the minimum detectable GDI at $\alpha = 0.005$ is roughly ± 0.15 , well above the actually-observed magnitudes. (ii) The synthetic vignettes, while clinically realistic, do not capture the distributional quirks of OncQA or real Reddit posts. (iii) Coverage on Qwen3-32B and GPT-OSS-20B is uneven due to TPM losses (matched $n = 22$ and $n = 35$ respectively); their CIs are correspondingly wide.

4.4 Baselines and Evaluation Methodology

The project-brief instruction to “*address any issues with baselines or evaluation methodology*” deserves explicit treatment. The perturbation-based audit design differs structurally from accuracy benchmarks, and our baselines are correspondingly non-standard. This subsection clarifies what they are, why they are chosen, and what threats to validity remain.

4.4.1 What the baseline is in this design

The primary baseline is *within-subject and within-model*: for each case c and each model f_θ , the Global-North condition $f_\theta(x_c^{(0)})$ is the reference point against which each Global-South perturbed condition $f_\theta(x_c^{(k)})$ is compared. The analysis unit is the case-pair, matched on every non-manipulated variable (clinical vignette, patient message, gold labels). This choice has three consequences that shape how the numbers in §4 should be read:

- Per-case variance in clinical complexity is eliminated as a confounder, because each case contributes a matched pair to the TSR / RCR / RCER computation. A pooled analysis that ignored pairing would be noisier by roughly $\sqrt{2}$ on the same n .
- Per-model base-rate differences in recommendation style (e.g. Qwen3-32B is substantially more conservative than GPT-4o-mini on VISIT in absolute terms; Table 5) cancel out, because the delta is computed within-model before any across-model aggregation.
- The design measures *disparity*, not *accuracy*. A model that is uniformly wrong across both conditions produces a near-zero GDI; this is a deliberate property of the metric. Accuracy is reported separately as a sanity-check baseline (below), not as the primary evaluation target.

4.4.2 What the baseline is not

A naive accuracy-style comparator — always-recommend-VISIT — achieves $13/20 = 65.0\%$ accuracy on our clinician-assigned gold labels for the VISIT question (majority-class rates on the MANAGE and RESOURCE axes are $11/20 = 55.0\%$ and $13/20 = 65.0\%$ respectively; source: `code/configs/cases.jsonl`). This is not a useful comparator for the audit question, because the question is not “*which model is accurate?*” but “*does identity substitution cause systematic recommendation shifts?*”. Under the primary matched-pair design (§4.1) the always-VISIT baseline has GDI identically zero by construction — it cannot exhibit disparity because it is a constant policy — so it provides no discriminating information. We nonetheless record per-model overall accuracy against the gold labels for completeness; none of the findings reported in §4 depend on it, and it is not used as a decision rule at any step of the pipeline.

4.4.3 External baselines from the literature

The TSR / RCR / RCER metric family was introduced by Gourabathina et al. [1] for tonal, gender, and syntactic perturbations on GPT-4. Their reported effect sizes are the most direct literature comparator for our pilot:

- **Gourabathina et al. (2025) [1]**. TSR ≈ 7 –9 pp and RCER shifts ≈ 7 pp for whitespace and colourful-language perturbations on GPT-4 evaluated on OncQA ($n = 61$), statistically significant at $p < 0.005$ under Bonferroni correction. These are *within-Global-North* identity shifts; no geographic-axis comparator is reported in that paper.
- **Omar et al. (2025) [17]**. Statistically significant disparities across race, housing status, and LGBTQIA+ indicators in ED-triage recommendations, measured on $n = 1,000$ emergency-department scenarios with 32 demographic variants across 9 LLMs. Reports per-axis effect sizes in the 5–15 pp range on triage decisions.

- **Pfohl et al. (2024) [9].** Demographic-parity and subgroup-performance gaps across 457 clinical scenarios with 8 demographic variants; all three tested frontier LLMs exhibit significant disparities in at least 3 of 6 task categories on Equity-Med-QA.

Our pilot’s per-model matched-pair GDI range (-0.062 to $+0.015$; Table 5) and per-question RCER shifts (the largest absolute per-question $|h|$ in the pilot is the -0.461 Cohen’s h on Qwen3-32B RESOURCE at $n = 13$; Table 6) sit below the effect-size bands of Gourabathina et al. and Omar et al. on comparable models. Our pooled matched-pair mean RCER shift is approximately one order of magnitude *smaller* than the Gourabathina et al. tonal-perturbation effects. Per DECISION #3 (logged in `code/decisions.md`), this pooled-null result is consistent with either (H1) post-training alignment having measurably suppressed geographic-axis bias, or (H2) per-model variance being masked by pooling at the pilot’s n ; the pilot is underpowered to discriminate the two, and the OncQA scaling experiment is pre-registered to do so (see §4 and `sections/results_reframe.tex`).

4.4.4 Power analysis and the significance threshold

Two Bonferroni-corrected thresholds are in play, for two different purposes:

- **Descriptive-pilot threshold:** $\alpha = 0.005$ applied at the region $\times$ question level, Bonferroni-corrected over the 9 conditions (3 Global-South regions $\times$ 3 outcome questions; see the `metrics.py` docstring and `runs/20260418T050306Z/power_analysis.json`). This is the threshold cited alongside Wilcoxon p -values in Table 5 and drives the required- n figures in Table 8 below.
- **H1-vs-H2 discrimination criterion:** the criterion for declaring per-model, per-question evidence *against* H1 uses 99.6% BCa bootstrap CIs, i.e. Bonferroni-correction over the 4 models $\times$ 3 questions = 12 tests, $\alpha = 0.05/12 \approx 0.0042$ (two-sided $z \approx 2.86$). This is the pre-registered criterion amended in `decisions.md` (AMENDMENT to DECISION #3) and stated in full in `sections/results_reframe.tex`: if any per-question RCER 99.6% BCa CI excludes zero in the OncQA $n = 61$ run, H2 is supported; if none does, H1 is the strictly stronger explanation. Per-model 95% BCa CIs on the per-case mean delta (Table 5) are descriptive and uncorrected; the Bonferroni adjustment applies only to the H1-vs-H2 discrimination test.

Using Cohen’s h [18] for paired proportions computed from each model’s observed ($\text{RCER}_{\text{North}}, \text{RCER}_{\text{South}}$) means, we determined the minimum matched-pair n required to detect each model’s observed pooled effect at power 0.80 under both thresholds, and at power 0.95 at the stricter threshold (Table 8). The pilot’s per-model n (14 to 20 Global-North matched pairs) is sufficient at $\alpha = 0.05$ to detect Qwen3-32B’s observed effect ($|h| = 0.183$, $n_{\text{req}} = 1$) but reaches neither model’s $\alpha = 0.005$ required sample. An OncQA $n = 61$ run clears the Qwen3-32B $\alpha = 0.005$ target ($n_{\text{req}} = 20$), approaches the Llama-3.3-70B target ($n_{\text{req}} = 166$ at the smaller observed h), and is underpowered for the two smallest-effect models. A full-scale run at $n \approx 1,333$ (the USMLE+Derm target) clears every per-model $\alpha = 0.005$ threshold at power 0.80 and clears the Qwen3-32B target at power 0.95 ($n_{\text{req}} = 174$); the remaining three models at power 0.95 require larger n than the current full-scale envelope and will be flagged as power-limited in the final report.

Table 8: Minimum matched-pair sample size to detect each model’s observed pooled RCER shift, computed from Cohen’s h in `runs/20260418T050306Z/power_analysis.json`. The $\alpha = 0.005$ columns are Bonferroni-corrected over 9 region $\times$ question conditions per `metrics.py` (descriptive-pilot threshold). Positive h indicates South $>$ North.

Model	Observed h	n ($\alpha=0.05$, 0.80)	n ($\alpha=0.005$, 0.80)	n ($\alpha=0.005$, 0.95)
GPT-4o-mini	+0.084	4	92	822
GPT-OSS-20B	−0.056	9	206	1,846
Llama-3.3-70B	−0.062	7	166	1,485
Qwen3-32B	−0.183	1	20	174

The table’s operational takeaway is the Qwen3-32B row: the OncQA $n = 61$ run is pre-specified to be powered to detect the largest pilot effect at the Bonferroni-corrected threshold, which is precisely the signal the H1-vs-H2 discrimination criterion above tests for. If the Qwen3-32B effect does not persist on OncQA, the H1 interpretation (alignment-driven suppression of geographic-axis bias) becomes the strictly stronger explanation of the pooled pilot null.

4.4.5 Threats to validity and mitigations

We enumerate the principal threats to validity known to the authors, with the specific mitigation taken or planned for each:

Gold-label provenance. Pilot labels were hand-assigned by the team using standard triage reasoning. A formal Emergency Severity Index v4 rubric [20] has been drafted (`code/docs/labeling_rubric`) and is pending an intra-team double-labelling exercise over the 20-case pilot set; Cohen’s κ [19] between the two independent labellers is **[TODO: pending]** (Agent-LABELS halted at the human-labeller step per `code/decisions.md` entry 2026-04-18 — Prompt 08). Full-scale datasets will use OncQA (clinician-validated by Chen et al. [21]), r/AskDocs (verified-physician-labelled per the construction in Gomes [22]), and USMLE+Derm (expert-annotated; Jin et al. [23], Johri et al. [10]).

Annotator reliability. The Llama-3.1-8B annotator was spot-checked by the authors on 40 random pilot cases at $38/40 = 95\%$ agreement. Formal Cohen’s κ against two independent human labellers on the same 40-case subset is **[TODO: pending]** (same Agent-LABELS blocker; see the rubric + `scripts/compute_kappa.py` harness already in place). Annotator heuristic-fallback rate in the final artefact is 0 after the serial re-annotation pass described in §3.3.

Name-phonological confounds. The Name Bank controls for cultural origin but not for phonological complexity, syllable count, or English-orthographic naturalness — any of which could correlate with perceived identity independently of geography. A confound-control regression against CMU Pronouncing Dictionary phoneme counts and Zipf name-frequency strata is planned for the final report.

Geographic-reference confounds. Substituting “Boston, MA” with “Lahore, Pakistan” changes not only perceived patient origin but also healthcare-system priors (e.g. drug availability, referral pathways). The NAME-only and GEO-only ablations (Table 4.2; planned, see §7) isolate these effects; the pilot reports the COMBINED condition only, which conflates them.

Model-inferred geography. Even when explicit identity signals are stripped, models may infer geography from writing style, dialect, or content choices [3]. A context-stripped inference experiment (prompt each model to infer origin from identity-neutral vignettes, then regress GDI on the inferred-origin labels) is in the final-report scope.

Temporal stability of model APIs. Provider-returned `model_id` is recorded per completion in `completions.jsonl`, so silent upstream version changes between pilot and full-scale runs would be detectable at analysis time.

Reproducibility. Every number in §4 traces to run 20260418T050306Z and to the specific artefact files (`summaries.json`, `power_analysis.json`, `annotated.jsonl`); the dataset and Name-Bank SHA-256 hashes are pinned in `manifest.json`; the cache keys every LLM call by SHA-256(model, prompt, seed, temperature), so re-running the pilot from scratch on the same config + seed reproduces the completions byte-identically. Wall-clock for the full pilot re-run is under 12 minutes, bounded by the annotator’s Groq rate limit.

5 Timeline and Progress

5.1 Completed Work

- **[DONE]** Literature review: 13 papers across 4 themes; accepted with no revisions.
- **[DONE]** Pilot dataset: 20 clinician-labelled vignettes (`code/configs/cases.jsonl`).
- **[DONE]** Geographic Name Bank v1.0: 150 entries, 25 per region, M/F balanced (`code/configs/name_bank`).
- **[DONE]** Perturbation engine: template-based, deterministic (SHA-256 keyed), 4 perturbation types.
- **[DONE]** Inference harness: OpenAI + Groq adapters, token-bucket rate-limiting, exponential-backoff retry, file-backed cache, idempotency keys, SHA-256 manifest.
- **[DONE]** LLM-as-annotator: Llama-3.1-8B JSON-constrained, three-layer fallback, serial re-annotation tool.
- **[DONE]** Pilot run 20260418T050306Z: 320 scheduled completions (268 successful, 52 lost to Groq 429s); all 320 rows annotated, 0 heuristic fallback after the serial re-annotation pass.
- **[DONE]** Pilot statistical analysis (Table 5–7): Wilcoxon, bootstrap CIs, per-question and per-region decomposition.

5.2 Work in Progress

- **[IN PROGRESS]** Per-model rate-limit buckets (currently per-provider). Target: ≤ 3 RPM on Qwen3-32B and GPT-OSS-20B to eliminate 429 losses in the full run. Draft patch in review.
- **[IN PROGRESS]** Dataset expansion: OncQA and r/AskDocs loaders drafted; access paperwork in progress.

- **[IN PROGRESS]** Add frontier models: GPT-5-family available through the existing OpenAI key; Claude/Gemini blocked on key acquisition.

5.3 Remaining Work

- **[PENDING]** Scale to real datasets: OncQA (gender-filtered), r/AskDocs, USMLE+Derm (CRAFT-MD).
- **[PENDING]** Run NAME-only and GEO-only perturbation ablations.
- **[PENDING]** Add Southeast Asia and MENA regions (already in the Name Bank) to the condition set.
- **[PENDING]** Multi-seed re-run (3 seeds) for the variance-critical experiments.
- **[PENDING]** Intersectional analysis: geography $\times$ gender, geography $\times$ severity.
- **[PENDING]** Model-inferred-geography experiment (prompt each model with identity-stripped vignettes and infer origin, regress GDI on inference).
- **[PENDING]** Final report and presentation.

6 Contribution and Team Coordination

6.1 Member Contributions

All five members contributed substantively. Ownership is assigned to ensure accountability, not exclusivity.

Member	Primary Contributions This Phase
Shawal Latif	Led the Name-Bank v1.0 construction (150 entries, 6 regions); coordinated cultural-representativeness review; co-authored <code>audit.perturb</code> ; wrote §5 and §6 of this report.
Usmar Haider	Built the inference harness (<code>audit.models</code>): OpenAI + Groq adapters, token-bucket, retry, cache, idempotency; debugged the Cloudflare-1010 User-Agent issue; wrote §3.1–3.2.
Taimoor Karim	Designed and calibrated the LLM-as-annotator (<code>audit.annotate</code>); wrote the statistical module (<code>audit.metrics</code> , including the hand-rolled Wilcoxon and bootstrap); produced Tables 5–7; wrote §4 and the Executive Summary.
Abdul Moeed Irshad	Constructed and gold-labelled the 20-vignette pilot dataset; wrote the dataset loader (<code>audit.data</code>); maintained manifest hashing; wrote §2 of this report.
Syed Muhammad Mujtaba	Co-authored <code>audit.perturb</code> ; built the CLI driver <code>audit.run</code> and <code>scripts/reannotate.py</code> ; diagnosed the 52 TPM losses and designed the serial re-annotation fix; wrote §3.3 and §7.

Table 9: Per-member primary contributions for the intermediate phase.

6.2 Coordination Practices

- Weekly 45-minute sync; written decisions logged in `decisions.md`.
- All code reviewed in pairs; the pipeline was bisected when the annotator started returning heuristic labels, which is how the TPM cause was isolated.
- Run directories are timestamped (`runs/<UTC>`) and contain a `manifest.json` capturing the cases and Name-Bank SHA-256 plus the resolved model specs, so every claim in §4 can be traced to a specific artefact.

7 Next Steps and Future Work

7.1 Immediate Next Steps (April 20 — May 15, 2026)

1. Ship the per-model token-bucket patch; re-run Qwen3-32B and GPT-OSS-20B to close the 52-completion gap. Expected outcome: complete $n = 20$ coverage on all four pilot models, enabling fair GDI comparison.
2. Load the first 61-case OncQA batch and re-run the pipeline unchanged. Sample-size goal: with the pilot’s observed per-case-delta variance (~ 0.15), Wilcoxon at $\alpha = 0.005$ needs roughly $n \approx 140$ matched pairs to detect a GDI of ± 0.05 ; OncQA + r/AskaDocs combined clears that threshold.
3. Add GPT-5-mini (available today via the existing OpenAI key) as a fifth model; target is to include Claude and Gemini once keys are in hand.
4. Run NAME-only and GEO-only ablations at the pilot scale to separate the contributions.
5. Produce the GDI heat-map (model $\times$ region) and the TSR decomposition plot.

Critical-path items. Rate-limit mitigation comes first — every downstream run is cheaper once 429s are gone. OncQA access is the next gate.

7.2 Anticipated Challenges

- **Cost of Claude/Gemini.** These require paid keys; we have budgeted for a small subset of each if approvals arrive in time.
- **Ceiling effects at the low-bias end.** GPT-4o-mini’s GDI is $+0.015$; at $n = 20$ we cannot discriminate this from 0. We will need $\sim n = 200$ matched pairs for a ± 0.02 CI, which the OncQA + r/AskaDocs combination just clears.
- **Annotator reliability at scale.** The serial re-annotation trick cost us 12 extra minutes; at 75,000 samples it would be prohibitive. A per-account Groq separation or swapping to a paid OpenAI annotator is the plan.

- **Cultural validity beyond the project team.** The Name Bank was peer-reviewed inside the group; for the final report we will run a lightweight survey (public Google Form) to collect external regional-representativeness scores.

7.3 Improvements for the Final Report

- **From pilot scale to publication scale:** 20 vignettes $\rightarrow \sim 1,540$ cases across OncQA + r/AskaDocs + USMLE+Derm.
- **From single axis to intersectional:** add gender and severity slices.
- **From detection to mechanism:** attention/activation analysis on the open-weights models (Llama-3.3-70B is a feasible target) to investigate *where* in the model the geographic signal influences outputs.
- **From a single metric to a ranking:** bootstrap-based ranks with pairwise significance.

8 Appendices

8.1 Code and Implementation

Repository layout (packaged with this submission under `code/`):

- `audit/data.py` — dataset loader + SHA-256 manifest hashing.
- `audit/perturb.py` — Name-Bank-driven perturbation engine (deterministic).
- `audit/models.py` — unified `generate(messages)` abstraction over OpenAI + Groq; token-bucket rate limiter; idempotent disk cache; stdlib transport.
- `audit/annotate.py` — Llama-3.1-8B annotator; three-layer JSON/regex/heuristic fallback.
- `audit/metrics.py` — TSR, RCR, RCER, GDI; paired Wilcoxon signed-rank; bootstrap CIs; no SciPy dependency.
- `audit/run.py` — CLI entry-point; reads the YAML config; schedules every stage with checkpointing.
- `scripts/reannotate.py` — serial re-annotation for heuristic-fallback records.
- `configs/pilot_oncqa.yaml`, `configs/cases.jsonl`, `configs/name_bank.json`.

Reproducing the pilot. With the two API keys in `.env`:

```
set -a; source .env; set +a
cd Module_3_Intermediate_Report/code
python3 -m audit.run --config configs/pilot_oncqa.yaml --seed 42 --parallelism 8
# If the run triggers 429s on Groq, re-annotate heuristic-fallback records:
python3 scripts/reannotate.py --run-dir runs/<latest-timestamp> --sleep 7
```

Each run directory contains `manifest.json`, `perturbed.jsonl`, `completions.jsonl`, `annotated.jsonl`, `summaries.json`, and a SHA-256-keyed response cache (`.cache/`).

8.2 Supporting Materials

- Run 20260418T050306Z is the exact pilot whose numbers appear in Tables 5–7. `manifest.json` locks the `cases.jsonl` SHA-256 and `name_bank.json` SHA-256.
- Every provider response is cached under the idempotency key; a second invocation with the same config + seed produces byte-identical metrics.
- All prompts (system, user, annotator) are embedded in `audit.run` and `audit.annotate` respectively and version-tracked with the code.
- Provider-returned `model_id` is recorded per completion, so silent upstream version bumps would be detected.

8.3 Sensitivity Analysis (Errors-Included View)

The full errors-included sensitivity narrative is in §4 under the subsection “Sensitivity analysis: the errors-included view,” together with the infrastructure caveat (40 of 52 failures preceded the per-model rate-limit refactor). The corresponding artefact is `runs/20260418T050306Z/summaries_errors`; its SHA-256 is logged in `code/decisions.md`. These values are not cited as findings of the pilot.

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